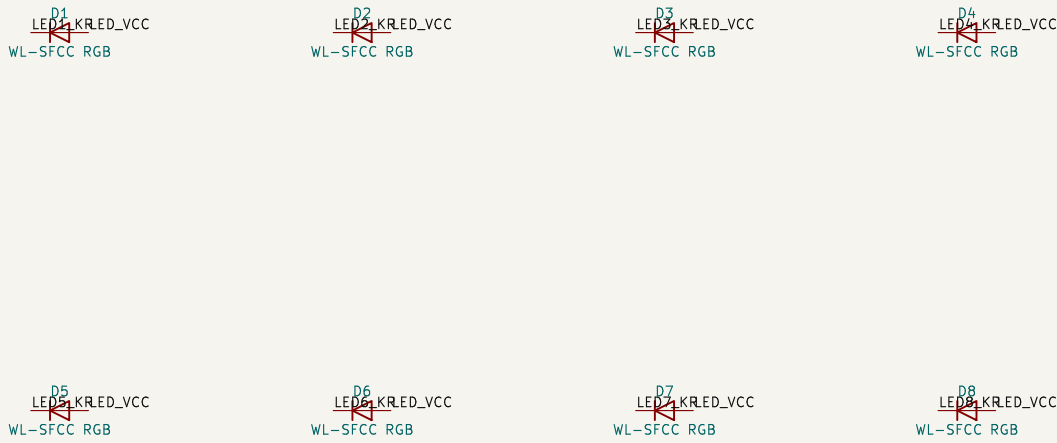


BLOCK 1 — LED MATRIX D1-D8 (8 × Würth WL-SFCC RGB common-anode) → cathodes route to south header H_S_1..32

BLOCK 4 — NTC THERMISTORS (TH1-TH4, Murata NCP15XH103J03RC, 10 kΩ ±5 %, B25/85 = 3380 K) → V_F-TSP thermal sensing of bonded LEDs

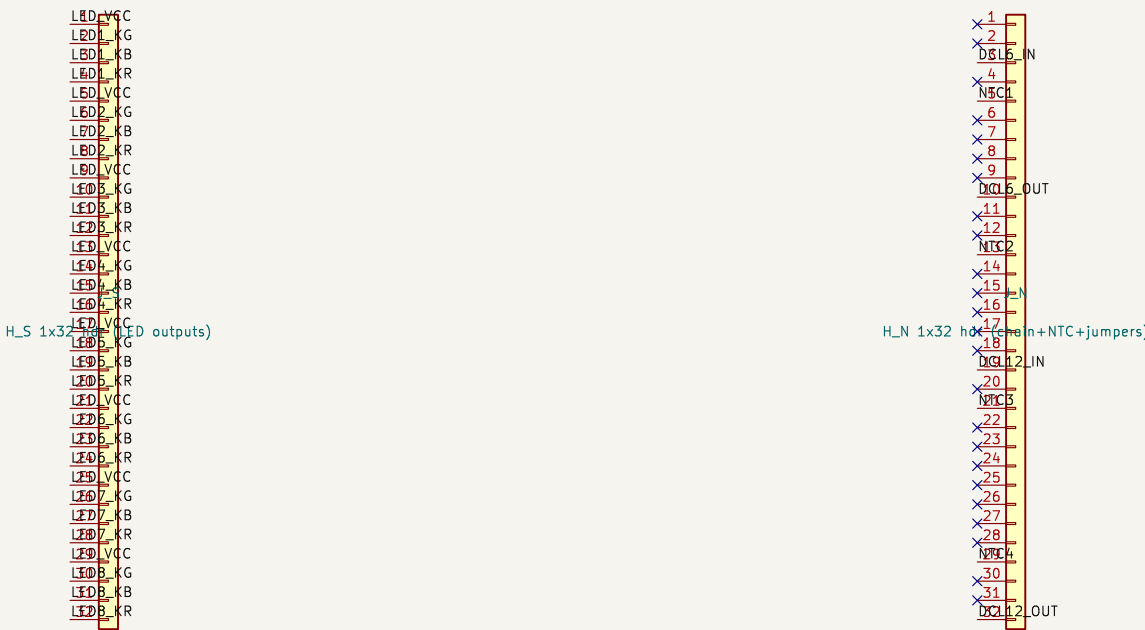


BLOCK 2 — DCL6 DAISY CHAIN (6 × WL-SFCC, A→K_R series) · K_G / K_B per LED isolated (probe-only) · endpoints route to H_N_3 / H_N_10

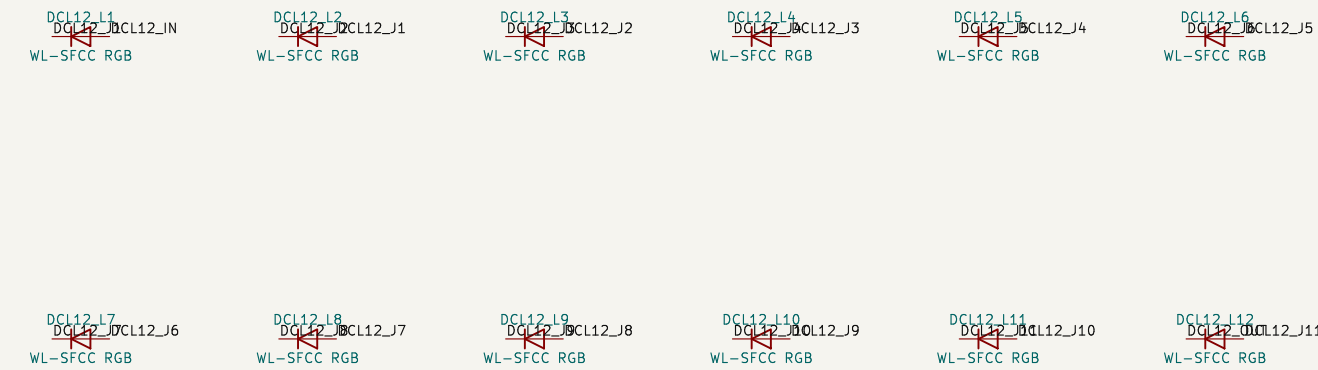
BLOCK 5 — EIS CALIBRATION (LCR-meter Nyquist reference: 100 Ω LOAD + OPEN (∞) + SHORT (0 Ω))



BLOCK 6 — HEADERS (2 × Würth WR-PHD 61304011121, 1×32 single-row male, 2.54 mm)



BLOCK 3 — DCL12 DAISY CHAIN (12 × WL-SFCC, A→K_R series) · endpoints route to H_N_19 / H_N_32



DESIGN NOTES — Test structures (bare ENIG pads, no electrical connections, not shown on schematic):

- DoE bond-pad array (36 × isolated 1 mm pads, 3.5 mm pitch)
- TLM ladders (3 widths × 7 fingers, 200/300/500/1000/2000/4000 μm gaps)
- Van der Pauw cloverleaves (W=1.0/0.5/0.25/0.15 mm × 4 contacts)
- Probe pads PP_* (46): 4-wire access to bonded LEDs
- Thermocouple pads TC1-4 (in-situ reflow temp)
- Optical fiducials FID1-4 (P&P alignment)

NET NAMING: LED_VCC = common anode bus · LEDn_K{R,G,B} = cathode of D1..D8 · DCL{6,12}_IN/OUT = chain endpoints · DCL{6,12}_k = chain junction k · DCLn_Lk_K{G,B} = isolated G/B cathodes (chain LEDs, probe-only) · NTCn = thermistor sense · EIS_LOAD_A/B = 100 Ω cal resistor · EIS_OPEN_A/B = open-circuit cal · EIS_SHORT = short-circuit cal · GND = common return

PCB-side notes: WL-SFCC LEDs depicted with single Devices:LED symbol per chip — K shown = K_R (in-chain cathode); K_G and K_B route to PCB pads via labels and reach the south header at pins (4k+2) and (4k+3).